



Insurance Risk & Claims Analysis

A centralized Power BI dashboard to unify policy and claims data — enabling data-driven decisions across the policyholder base.

The Business Challenge



The Problem

Policy and claims data are **scattered across multiple sources**, making it difficult for stakeholders to track performance and identify trends.

The Solution

A **centralized, interactive Power BI dashboard** providing a clear, unified overview of policyholder behavior and claim patterns.

Key Performance Indicators

Total Policies

Measure the size of the active customer base.

Total Claim Amount

Track the overall financial impact of claims.

Claim Frequency

Analyse how often claims are being made.

Avg Claim Amount

Assess claim severity and potential risk exposure.

Gender-wise Policies

Understand customer distribution for better segmentation.

Two Core Dynamic Measures

All visualizations are built around two key dynamic measures that provide the foundation to compare, filter, and segment data effectively.

Total Claim Amount

Tracks the financial weight of claims across every dimension — geography, demographics, and vehicle data.

Total Policies

Measures policyholder volume to reveal distribution patterns and segment-level risk concentration.

Vehicle-Based Visualizations



By Car Use

Donut Chart — Analyse policy distribution and claim amounts by car usage: personal vs. commercial.



By Car Make

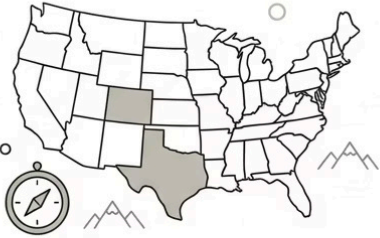
Bar Chart — Identify which car brands carry higher policies and claims, highlighting brand-based risk.



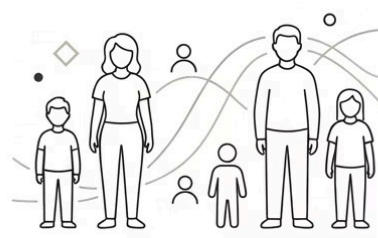
By Car Year

Area Chart — Analyse how a vehicle's age (year of manufacture) impacts policy counts and claim amounts.

Geographic & Demographic Insights



COVERAGE ZONE:
Geographic
Regional Risk



AGE GROUP:
Which brackets
file more claims



EDUCATION:
Policy adoption &
claims correlation



GENDER:
Customer distribution
for segmentation

Coverage Zone

Donut Chart — Evaluate policies and claims by geographic zone for regional risk analysis.

Age Group

Histogram — Assess policyholder age distribution and identify high-claim age brackets.

By Education

Pie Chart — Understand how education levels correlate with insurance adoption and claims.

Household & Behavioral Analysis

By Kids Driving

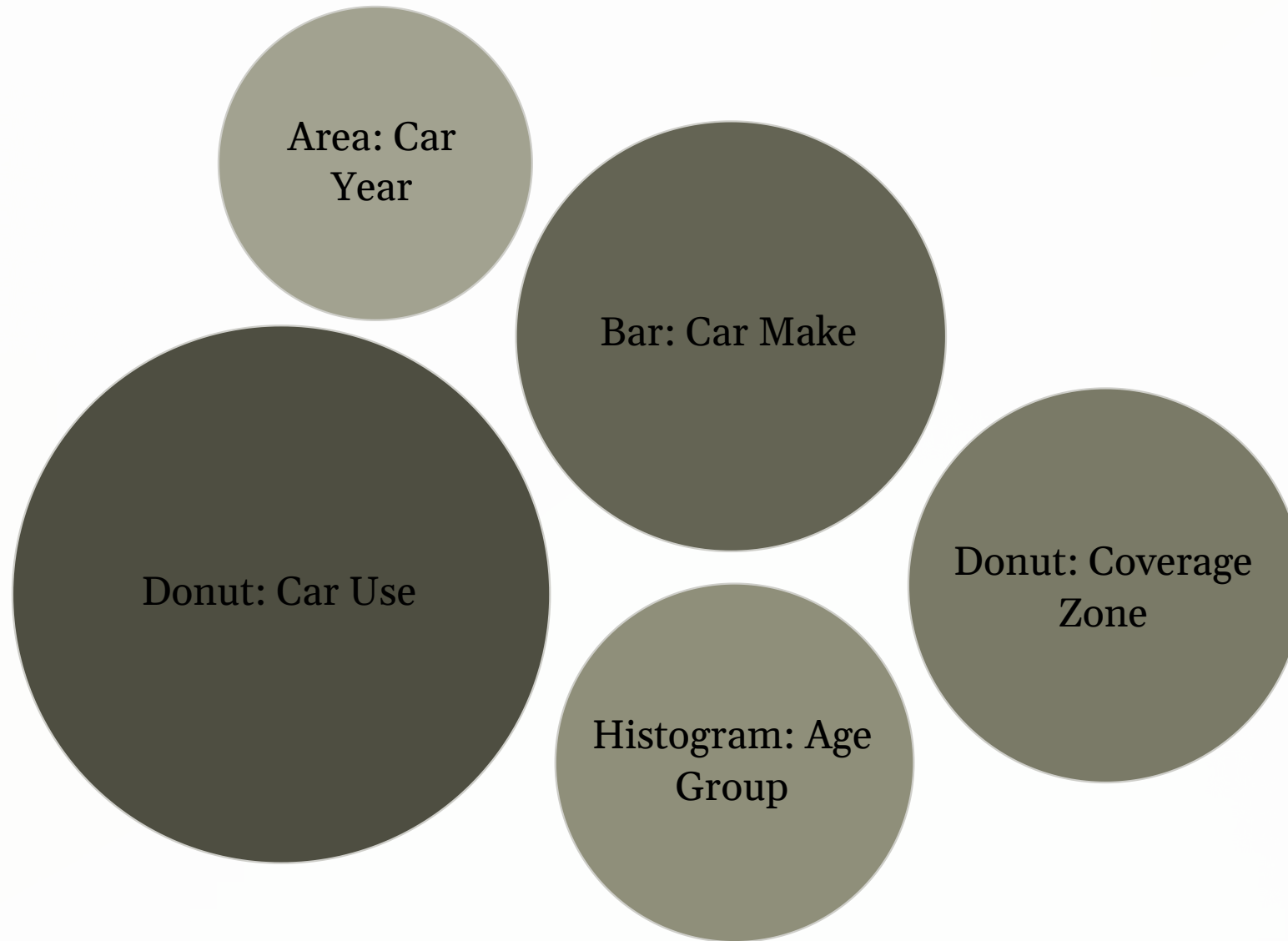
Ribbon Chart — Compare the impact of young drivers in households on policy count and claim amounts. Reveals how household composition shifts risk profiles.

By Education & Marital Status

Matrix Heat Grid — Explore the combined effect of education and marital status on policies and claims, surfacing distinct customer profiles.



Full Visualization Suite



Each chart type is selected to best reveal patterns within its data dimension — from geographic risk to demographic behavior.

Business Impact

Identify Risk Factors

Spot high-risk segments by vehicle type, age group, geography, and household profile.

Understand Customer Segments

Leverage demographic and behavioral data to tailor policy offerings and targeting.

Optimize Policy Decisions

Make data-driven adjustments to pricing, coverage, and underwriting strategies.



Summary

01

Centralize Data

Unify scattered policy and claims sources into one Power BI environment.

03

Explore 8 Visualizations

Drill into vehicle, geographic, demographic, and behavioral dimensions.

02

Track 5 KPIs

Monitor total policies, claim amounts, frequency, severity, and gender distribution.

04

Drive Decisions

Empower stakeholders with real-time, interactive insights to reduce risk and grow the business.